Question 1

a. Limitations of the on-premise approach.

A fully on-premise GPU facility would require a massive capital outlay in servers, storage, networking and the supporting infrastructure. You'd need to buy out the capacity for exam periods even if a huge proportion of it would be idle during the semester breaks. You'd also be very slow to scale up as you'd have to buy and install new equipment. GPUs also move on so quickly and you'd be left with the full burden of power, cooling, maintenance, physical security and hardware replacement.

b. How cloud computing addresses these limitations.

SACF can increase its use of resources during examination periods or as part of extensive research projects and decrease them when demand drops. This ability to scale up and down means there's no need to buy hardware just to meet occasional peaks in demand. With a pay-per-use arrangement, spending is directed towards the resources that are actually being used, and since resources are pooled, various faculties and research groups can share the available capacity. AWS also enables SACF to gain access to more modern computing resources without the University having to buy and replace each generation of GPU hardware. However, the University would still have to keep an eye on both usage and spending in order not to pay for resources that are idle.

c. Impact of cloud adoption.

For AI research, faster access to GPUs and high-performance storage can reduce waiting times and help researchers run more demanding experiments. For teaching and learning, students can use consistent, research-grade computing environments for practical work, including when demand rises during examinations. A shared SACF environment can support interdisciplinary collaboration by enabling teams from different faculties to use common resources while applying suitable controls to their data and projects. Students who gain experience with cloud-based AI infrastructure and collaborative workloads can develop practical skills that improve their

preparation for employment. These benefits support Sunway's aims in AI research, digital-first teaching, collaboration, and future-ready graduate development.

d. Risk and mitigation.

A significant risk is unexpected cloud expenditure if GPU instances or storage remain active when they are no longer needed. SACF should set budgets and spending alerts for each faculty or project, apply quotas or approval procedures to expensive GPU resources, and review usage regularly. Automatically stopping idle teaching environments would further reduce unnecessary costs.

Question 2

a. Comparison of the deployment options.

Scalability: Option A is limited to the capacity of the purchased cluster, and expansion requires new equipment. Option B can scale AWS resources as workloads change. Option C can use AWS to provide additional capacity when demand exceeds the on-premise resources.

Security and compliance: Option A gives the University direct control over its infrastructure and data, but Sunway must provide and maintain the necessary security. Option B provides AWS security capabilities, but Sunway remains responsible for correct configuration, access control, and data handling. Option C can keep selected sensitive workloads on university-controlled infrastructure and use AWS for other workloads, but securing and integrating both environments is more complex.

Cost: Option A requires high upfront capital expenditure as well as ongoing maintenance, power, and cooling costs. Option B has lower initial infrastructure costs and uses operational expenditure, but spending must be monitored. Option C requires some on-premise investment as well as AWS spending, but may avoid purchasing enough local hardware to cover peak demand.

Performance: Option A can provide predictable performance for workloads that fit the cluster, but its capacity is fixed. Option B provides access to specialised computing resources, though performance depends on the resources selected and their availability. Option C combines local capacity with the ability to use AWS for additional workloads; transferring data between the environments may add delay.

Manageability: Option A requires the University to manage hardware, upgrades, maintenance, and availability. Option B avoids managing physical cloud hardware but requires staff to manage cloud configuration, access, and costs. Option C requires staff to operate and secure both environments and manage their integration, making it the most complex option to administer.

b. Recommended deployment model.

Option C, a hybrid architecture, best balances the stated requirements if the on-premise component is kept small and reserved for workloads that genuinely require university control. AWS can provide elastic capacity for variable AI and teaching workloads, helping Sunway limit capital expenditure and adopt new technologies without waiting to buy hardware. Selected sensitive data or workloads can remain in the university-controlled environment. Because availability is critical during examinations and research deadlines, SACF should plan capacity in advance and test backup and recovery arrangements. The University must also account for the extra management complexity of operating both environments.

c. Recommended cloud service model.

IaaS is the most suitable primary service model for SACF because the facility needs configurable GPU computing, storage, networking, and research environments. It gives faculties and researchers more control over the operating systems and software needed for different workloads, without requiring Sunway to own all the underlying infrastructure. PaaS can simplify application development, but may offer less control than some specialised research workloads require. SaaS provides ready-to-use applications rather than the flexible computing infrastructure SACF is intended to provide. PaaS and SaaS may be useful for particular activities, but are less suitable as SACF’s main service model.